

Microsoft Certified Azure AI Engineer Associate [AI-102]

Cheat Sheet

Quick Bytes for you before the exam!

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AI-102: CheatSheet Index -

Topics Names	Page No
Plan and manage an Azure AI solution	
Azure AI Services - Overview	3
Azure AI Services - Planning, creating and deployment	9
Manage, monitor, and secure an Azure AI service	12
Implement content moderation solutions	
Azure AI Content Safety - Overview	16
Azure AI Content Safety - Text Moderation	16
Azure AI Content Safety - Image Moderation	16
Implement computer vision solutions	
Choose between image classification and object detection models	17
Implement custom computer vision models by using Azure AI Vision	18
Analyse images	19
Analyze videos	20
Implement natural language processing solutions	
Analyze text by using Azure AI Language	21
Process speech by using Azure AI Speech	22
Azure AI Translator and Speech Translation Services	22
Create a custom question-answering solution by using Azure AI-Language	23
Implement knowledge mining and document intelligence solutions	
Implement an Azure AI Search solution	26
Implement an Azure AI Document Intelligence solution	27
Implement generative AI solutions	
Use Azure OpenAI Service to generate content	28
Optimize generative AI	29

Plan and manage an Azure AI solution

Azure AI Services - Overview

Azure AI Vision Overview

Azure AI Vision is a comprehensive service within Microsoft Azure's suite of artificial intelligence tools that offers advanced image processing, text recognition, and facial detection capabilities. This service enables developers to integrate computer vision features into their applications without requiring expertise in machine learning, making it accessible for a broad range of use cases in different industries.

Key Features of Azure AI Vision

1. Image Analysis

- Automatic Tagging: Uses a database of 10,000+ concepts to auto-tag and classify images, making it easier to identify objects, create captions, and gain insights.
- Customizable Options: Flexible for specific business needs, allowing custom solutions.

2. Optical Character Recognition (OCR)

- Text Extraction: This tool extracts text from printed or handwritten documents across various languages, making it useful for document digitization and data entry.
- Supports Multiple Styles: Accurate recognition across diverse writing styles and languages.

3. Facial Recognition

- Face Detection and Verification enable apps to identify and verify faces for security, customer ID, and personalized experiences.
- Ethical Use: Built with Microsoft's responsible AI guidelines to ensure ethical deployment.

4. Spatial Analysis

- Real-Time Movement Tracking: Tracks presence and movement within physical spaces, useful for behavior insights, security, and occupancy control.
- Ideal for Physical Locations: Applicable in retail, workplaces, and similar spaces needing spatial data.

Azure AI Vision Pricing Overview

Azure AI Vision offers a flexible pricing approach that benefits both new users and ongoing customers:

- **Free Trial:** New users get a \$200 credit for 30 days, allowing them to try Azure AI Vision and other services at no cost.
- **Pay-as-You-Go:** After using the free credit, users pay only for what they use beyond the free limits, with certain AI features remaining free for 12 months.

Azure NLP Overview

Azure's Natural Language Processing (NLP) enables apps to interpret and analyze language data with minimal effort. It includes tools for entity recognition, sentiment analysis, summarization, conversational AI customization, and health data processing, all powered by advanced transformer models.

Key Features

1. **Entity and Data Privacy:** Detects and redacts personal data, supporting privacy compliance.
2. **Sentiment & Language Analysis:** Identifies language and emotional tone for better customer insights.
3. **Summarization:** Extracts key points from long content, making it easy to digest information.
4. **Conversational AI:** Builds responsive chatbots with multilingual support and intent recognition.
5. **Health Data Insights:** Converts unstructured medical text into structured data for better patient care.

Use Cases

- **Customer Service:** Automated support via chatbots.
- **Compliance:** Redacts personal data for privacy adherence.
- **Market Insights:** Analyzes brand sentiment and trends.
- **Healthcare:** Extracts actionable data from medical records

Azure AI Speech

What is Azure AI Speech?

Azure AI Speech is a comprehensive, cloud-based service that enables developers to create high-quality voice-enabled applications with features such as speech-to-text, text-to-speech, and speaker recognition. It supports real-time transcription, multilingual translation, and custom voice models, streamlining app creation for voice-driven experiences.

Key Features of Azure AI Speech

1. **Speech-to-Text:** Transcribes speech to text in 100+ languages, aiding customer insights and meeting transcription.
2. **Text-to-Speech:** Produces natural-sounding voice outputs, ideal for voice assistants and accessibility enhancements.
3. **Speech Translation:** Real-time translation for over 30 languages, with customizations for specific terms and contexts.
4. **Speaker Recognition:** Identifies or verifies speakers in conversations for added personalization or security.
5. **Custom Commands:** Enables voice commands for hands-free scenarios, like IoT devices and virtual assistants.
6. **Customizable Models:** Offers options to create custom voices, add unique vocabulary, or develop bespoke models.
7. **Secure and Compliant:** Prioritizes data security, with no logging of user speech data during processing.
8. **Flexible Deployment Options:** Deploys either in the cloud or on the edge, including container support for scalability.

Pricing Structure

- **Pay-as-You-Go:** Charges are based on usage with no upfront costs.
 - **Speech-to-Text and Translation:** Billed per hour of transcribed/translated audio.
 - **Text-to-Speech:** Billed per number of characters converted to audio.
 - **Speaker Recognition:** Billed per transaction for speaker identification or verification.

Generative AI

What is Generative AI?

Generative AI, a branch of Artificial Intelligence (AI), creates original content like text, images, or code by analyzing context and using advanced algorithms. Examples include tools like Microsoft Copilot, which enhances experiences by understanding user inputs and generating coherent outputs across various formats.

Key Features

- **Natural Language Processing (NLP):** Accurately interprets and responds to human language.
- **Content Generation:** Produces diverse outputs such as text, images, and code.
- **Contextual Understanding:** Understands context for more relevant responses.
- **Adaptability:** Learned from interactions to improve accuracy over time.

Use Cases

- **Conversational Assistants:** Powers chatbots for natural interactions.
- **Creative Content Generation:** Creates artwork, music, and literature.
- **Code Generation:** Helps developers with code writing and debugging.
- **Personalized Recommendations:** Tailor's suggestions to user preferences.
- **Content Automation:** Generates multimedia content for campaigns and platforms.

Azure Document Intelligence Overview

Azure Document Intelligence is an AI-powered solution designed to extract, organize, and process text, tables, and structures from various documents efficiently. It leverages pre-built and custom models to transform unstructured documents into structured, actionable data, either on-premises or in the cloud.

Key Features

- **Simple Text Extraction:** Extracts text, key-value pairs, and tables automatically without the need for manual labeling.
- **Customized Results:** Offers custom extraction for specific document formats, enhanced by human feedback for accuracy.
- **Flexible Deployment:** Deploys in the cloud, on the edge, or in hybrid environments with support for Azure Kubernetes Service (AKS) and Azure Container Instances.
- **Built-in Security:** Ensures enterprise-grade security and privacy compliance for data and models.

Use Cases

- **Document Processing:** Automates data extraction from receipts, invoices, and forms.
- **Data Analysis:** Organizes extracted information for integration with data visualization and analytics tools.
- **Business Automation:** Enhances workflows by integrating extracted data into backend systems for tasks like claims and invoice processing.
- **Intelligent Search Indexes:** Enables precise information retrieval by indexing document content.

Pricing Options

- Azure Document Intelligence offers both free and standard pricing plans, allowing users to pay based on the volume of data processed. This cost-effective approach reduces the manual effort required for document processing.
- Azure Document Intelligence simplifies document handling, boosts productivity, and integrates smoothly with AI-driven workflows for smarter data management.

Knowledge Mining Solution Overview

Knowledge mining enables organizations to extract insights from vast data sources, revealing valuable patterns and information that would typically remain hidden. This solution comprises three main phases: ingest, enrich, and explore, using Azure Cognitive Search and Azure AI services for powerful data transformation and access.

Key Phases and Features

→ Ingest

- ◆ **Data Source Integration:** Connect to Azure sources (e.g., Blob Storage, Cosmos DB) and third-party sources with dedicated connectors.
- ◆ **Format Compatibility:** Extract content from PDFs, Microsoft Word, PowerPoint, and more, turning diverse file types into structured data.

→ Enrich

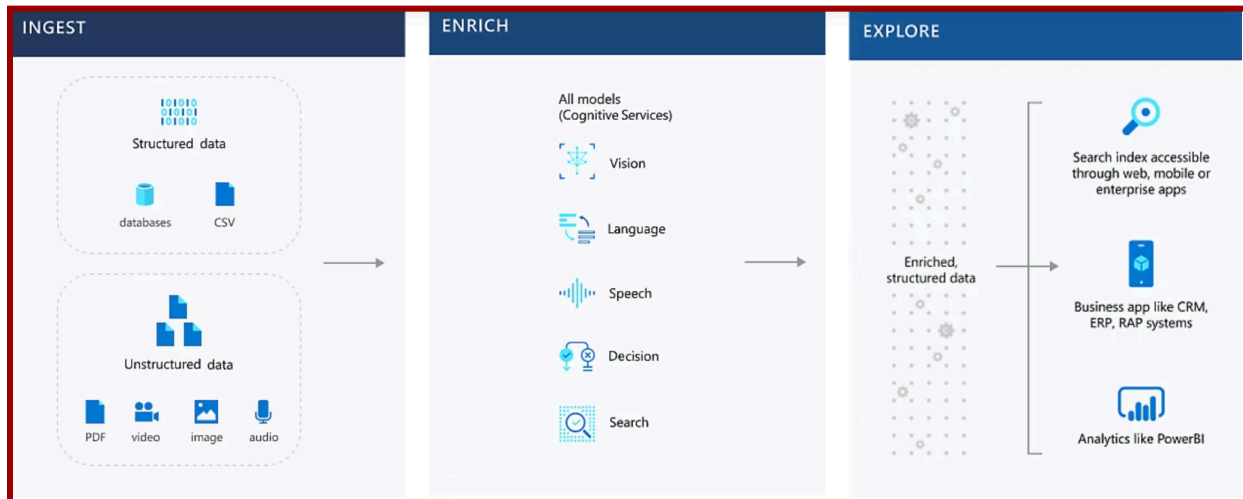
- ◆ **AI Enrichment:** Apply AI models (language, vision, decision) to extract key details, uncover patterns, and deepen data understanding.
- ◆ **Machine Learning Integration:** Use Azure Machine Learning for custom skills tailored to specific business or industry needs.

→ Explore

- ◆ **Enhanced Search:** Access enriched data through AI-powered Azure Cognitive Search, allowing intuitive data exploration.
- ◆ **Bot and Application Access:** Facilitate data discovery with bots, applications, and visualization tools.

Use Cases

- **Content Research:** Streamlines document review and data extraction.
- **Compliance and Risk Management:** Helps identify key information across legal and audit documents.
- **Customer Support:** Enhances response accuracy by finding the right information quickly.
- **Digital Asset Management:** Simplifies locating unstructured data within large repositories.



Source: Microsoft Documentation

Pricing

- Knowledge mining solutions in Azure are typically usage-based, with costs depending on data volume, complexity of AI models, and API usage. Pricing flexibility allows businesses to scale based on specific needs.
- **Azure Knowledge Mining** transforms large data pools into structured insights, enhancing efficiency, compliance, and customer satisfaction with scalable, cost-effective AI.

Azure AI Services - Planning, creating and deployment

Responsible AI in Azure Machine Learning

Responsible AI, also known as Ethical AI, emphasizes aligning AI with ethical principles to respect human rights and values and to mitigate unintended biases and potential harm. Azure Machine Learning (Azure ML) incorporates Responsible AI principles to ensure its AI systems are fair, safe, and accountable.

Key Principles and Features

- **Equity and Inclusion:** Azure ML uses fairness assessment tools to evaluate and mitigate biases, ensuring fair treatment across diverse groups.
- **Reliability and Safety:** By implementing error analysis tools, Azure ML supports secure and consistent AI system performance, reducing risks and operational errors.
- **Privacy and Security:** Azure ML enforces rigorous data protection standards to maintain user privacy, integrating robust security protocols.
- **Transparency:** Azure ML provides interpretability tools, helping users understand model predictions, which fosters trust and transparency in AI.
- **Accessibility:** With an emphasis on inclusivity, Azure ML enables the development of accessible models, broadening AI access for all users.
- **Accountability:** Azure ML upholds responsible AI standards, encouraging accountability and transparency throughout the AI development lifecycle and helping ensure stakeholders are accountable for AI outcomes.

Azure Machine Learning's Responsible AI framework promotes ethical, transparent, and secure AI practices, supporting organizations in deploying trustworthy AI solutions.

Adopt Responsible and Trusted AI Principles

Microsoft's Responsible AI principles ensure that AI systems are fair, safe, inclusive, transparent, and accountable.

Six Key Principles

- **Fairness:** Treat all users equitably, avoiding discrimination by ensuring consistent recommendations.
- **Reliability and Safety:** Design AI to operate reliably and securely across different scenarios.
- **Privacy and Security:** Safeguard personal data, implementing strong defences against unauthorized access.
- **Inclusiveness:** Create accessible AI for diverse users, addressing potential exclusion barriers.

- **Transparency:** Make AI decisions understandable, clearly communicating how the system works.
- **Accountability:** Hold developers accountable for upholding Responsible AI standards to foster trust.

Microsoft Azure Tools for Responsible AI

- **Content Safety:** Filters harmful content (text/images) with analysis for categories like hate and violence.
- **Transparency Notes:** Offers guidance for responsible AI use across Azure AI services.
- **Responsible AI Dashboard:** Provides tools like data analysis, fairness and error assessments, model interpretability, and “what-if” scenarios for a better understanding of model outcomes.

By embedding these principles and utilizing Azure’s tools, organizations can create trustworthy and ethical AI systems.

Create an Azure AI Resource - Demo

To create an Azure OpenAI resource in the Azure portal, follow these steps:

1. **Identify the Resource:** Sign in to the Azure portal using your Azure subscription and search for "Azure OpenAI" in the *Create a Resource* section.
2. **Fill in Basics:**
 - **Subscription:** Choose your Azure subscription.
 - **Resource Group:** Create or select a resource group to house the resource.
 - **Region:** Choose the location for your resource.
 - **Name:** Assign a descriptive name to your OpenAI resource.
 - **Pricing Tier:** Select the pricing tier (currently only Standard).
3. **Network Security Options:** Configure access to the resource:
 - **Allow All Networks:** Open access to all networks.
 - **Allow Selected Networks:** Specify virtual networks and subnets for restricted access.
 - **Disable Network Access:** Use private endpoints for exclusive access.
4. **Review & Create:** Review your settings and select "Create."

Endpoint for a Service

Azure provides an endpoint for each AI resource to facilitate API interactions. When a model is deployed, this endpoint allows developers to interact with the model through REST API calls, providing seamless integration with applications.

Integrate Azure AI Services into CI/CD

To streamline deployments, integrate Azure AI services into a CI/CD pipeline:

1. **Create Resource Templates:** Use ARM templates to configure infrastructure and resources, including AI services, enabling automation across environments.
2. **Pipeline Configuration:** Define stages in tools like Azure DevOps to manage resource creation, model deployment, and endpoint exposure, ensuring updates flow smoothly from development to production.

Plan and Implement a Container Deployment

Deploying AI models as containers allows for scalability and isolation:

1. **Container Setup:** Use Docker to package the AI model with dependencies.
2. **Azure Kubernetes Service (AKS):** Host containers on AKS for high availability and scaling.
3. **Deployment Management:** Implement CI/CD to update containers with newer model versions, enabling agile deployment and quick scaling.

Manage, monitor, and secure an Azure AI service

Configuring Diagnostic Logging

1. Access Diagnostic Settings:

- In the **Azure portal**, locate the desired Azure AI service (like **Speech Services**).
- Under **Monitoring**, select **Diagnostic settings**.

2. Create a Diagnostic Setting:

- Choose **+ Add diagnostic setting** (if no settings exist).
- Name the setting, then enable **Archive to a storage account** and **Send to Log Analytics** options.

3. Configure Storage and Log Analytics:

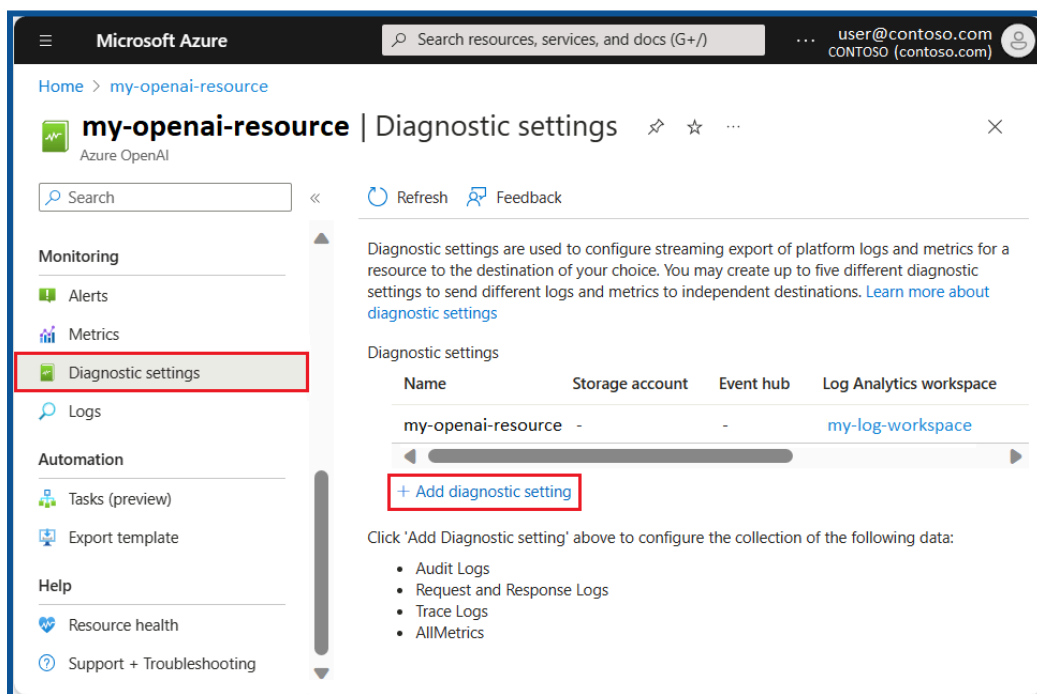
- Select an **Azure Storage account** and an **OMS workspace** to hold the logs. (If these don't exist, follow the prompts to create them.)

4. Choose Log Categories and Set Retention:

- Select log categories (**Audit**, **RequestResponse**, and **AllMetrics**).
- Set a **retention period** for the logs (entering 0 keeps logs indefinitely).

5. Save and Apply Settings:

- Click **Save**. It may take up to two hours for the logs to start appearing in queries.



Source: Microsoft Documentation

Monitoring Azure AI Resources with Azure Monitor

1. **Overview:** Azure Monitor provides visibility into availability, performance, and resilience by collecting metrics and logs from Azure AI resources. You can access monitoring via the Azure portal, CLI, PowerShell, or APIs.
2. **Dashboards:** Prebuilt dashboards for Azure AI resources show key metrics like **HTTP Requests**, **Token Usage**, **PTU Utilization**, and **Fine-tuning**. Access these from the **Overview** pane.
3. **Data Collection:** Azure Monitor gathers **Activity Logs**, **Resource Logs**, and **Platform Metrics**. Logs can be routed to storage locations like **Azure Storage** or **Event Hubs** via diagnostic settings.
4. **Data Storage:** Metrics go to the **metrics database**, logs to **Log Analytics**, and **Activity Logs** to a separate store. Data can be visualized in **Power BI** or **Grafana**.
5. **Diagnostic Settings:** Set diagnostic configurations to capture desired log categories and route data. Export data to **Log Analytics** to analyze with **Kusto Query Language (KQL)**.
6. **Export & Analysis:** Use **Metrics API** or **Logs API** for data export, or query within Log Analytics with **KQL** for resource-specific or cross-resource insights.

Managing Azure Storage Account Keys

Protect Access Keys

- **Azure Storage Account Keys** provide full access to your data. To secure them:
 - Use **Azure Key Vault** for management and rotation.
 - Avoid hard-coding keys or saving them in plain text.
 - Rotate keys periodically to minimize security risks.
 - Disable **Shared Key** authorization when not needed.
 - Use **Managed Identities** or **Microsoft Entra ID** for superior security, especially in Azure applications.

Use Azure Key Vault

- **Azure Key Vault** offers secure storage and management of your access keys:
 - Avoid storing keys in the application code.
 - **Azure Key Vault** allows easy key rotation without disrupting services.
 - Supports **PowerShell** and **CLI** for integration.

Manually Rotate Keys

- **Periodically rotate keys for better security:**
 - Use the **Azure Portal**, **PowerShell**, or **CLI** to regenerate keys.
 - Avoid using both keys simultaneously in different parts of the application.
 - Update **connection strings** after key regeneration.

Create a Key Expiration Policy

- **Set a Key Expiration Policy to monitor and ensure key rotation:**
 - Use **Azure Policy** to enforce key expiration checks.
 - This policy ensures that keys are rotated within the recommended time frame.

Monitor Compliance

- **Use Azure Policy to monitor whether keys are rotated on schedule:**
 - Track compliance through the **Azure Policy dashboard**.
 - Ensure that storage accounts meet security and key expiration standards.

Managing Azure AI Costs

- **Estimate Costs:** Use the **Azure Pricing Calculator** for Azure OpenAI cost projections.
- **Billing:** Charges are based on token usage (input and output). **Fine-tuned models** incur costs for training, hosting, and inference hours.
- **Extra Costs:** Enabling **Azure Monitor** or alerting can add additional fees. **Prepayment credits** can cover most charges, except Marketplace services.
- **Error Handling:** You'll be charged for **processing** even if the service returns error codes like **400** or **408**.
- **Cost Monitoring:** Use **Cost Analysis** to track and manage spending. Set **budgets** and receive **alerts** for overspending.
- **Exporting Data:** Export cost data to **Azure Storage** for further analysis.

The screenshot displays the Azure Cost Management interface. On the left, the navigation pane shows 'Reporting + analytics' and 'Cost analysis' highlighted. The main content area is titled 'C+L Vendor Managed Services Subscriptions / BAH-OpenAI-RG1 (change)'. It features a 'New tab' button, 'Refresh', 'Try preview', and 'Go to resource group' options. Below this, there's a 'Recent' section with 'All views' and 'Settings' buttons. The 'Recommended' section shows four tiles: 'Accumulated costs' (highlighted), 'Resources', 'Daily costs', and 'Services'. At the bottom, a table lists 'Smart views' with columns for Name, Group by, Date range, and Filters.

Name ↑	Group by ↓↑	Date range ↓↑	Filters ↓↑
Smart views			
Reservations	Reservations	This month	--
Resources	Resources	This month	--
Services	Services	This month	--
Customizable views			

Source: Microsoft Documentation

Authenticating Requests to Azure AI Services

Authentication Methods

There are three primary methods to authenticate requests to Azure AI services:

→ Authenticate with a Single-Service Resource Key

- ◆ Each Azure AI service provides a **resource key**.
- ◆ In the **Azure portal**, navigate to your resource and find the **Keys & Endpoint** section.
- ◆ Copy either **KEY1** or **KEY2** to authenticate your API requests.
- ◆ Include the key in the **Ocp-Apim-Subscription-Key** header when making requests.

→ Authenticate with a Multi-Service Resource Key

- ◆ For services that require authentication for multiple resources, use the **multi-service resource key**.
- ◆ You also need the **Ocp-Apim-Subscription-Region** header for services like **Azure AI Translator** to specify the region.

→ Authenticate with an Access Token (Authorization Header)

- ◆ If using **OAuth2** or other token-based authentication, use the **Authorization** header.
- ◆ Format: **Bearer <TOKEN>**.

Best Practices

- Use **two keys** for secure key rotation without service disruptions.
- Regularly **rotate** your keys to maintain security.

Implement content moderation solutions

Azure AI Content Safety - Overview

- **Purpose:** Azure AI Content Safety is a tool designed to detect harmful or offensive content across applications. It moderates both user-generated and AI-generated content.
- **APIs for Detection:** Provides text and image moderation APIs to identify and filter harmful content.
- **Content Safety Studio:** An interactive platform allowing users to explore moderation tools, try sample code, and create customized content safety workflows. Supports profane content flagging and custom blocklists.
- **Use Cases:** Used in generative AI services, e-commerce marketplaces, social platforms, and educational tools to ensure safe and compliant user experiences.

Azure AI Content Safety - Text Moderation

- **Text Analysis API:** Scans text for harmful content like violence, hate speech, or self-harm with multi-level severity options.
- **Prompt Shield:** Analyzes user input for security risks, specifically for attacks on large language models.
- **Groundedness Detection:** Checks if AI-generated text aligns with user-provided sources.
- **Protected Material Detection:** Flags sensitive material like copyrighted text (e.g., song lyrics or articles).
- **Custom Categories API:** Allows defining and scanning for custom categories, tailored to unique harmful content patterns.

Azure AI Content Safety - Image Moderation

- **Image Analysis API:** Detects inappropriate content in images (sexual, violent, hateful) across severity levels.
- **Configurable Filters:** Users can set sensitivity levels for image moderation, control blocklists, and export settings directly to applications.
- **Content Safety Studio Features:** Enables interactive image moderation testing, including response tracking and sensitivity adjustments for precise filtering.

Implement computer vision solutions

Choose between image classification and object detection models

Choosing between **image classification** and **object detection** models depends largely on your project goals, the complexity of the objects within images, and the level of detail required.

1. Use Image Classification When You Need Broad Categorization

- **Single Dominant Object:** If each image contains one main subject (such as a type of animal or a piece of equipment), image classification is sufficient. It can answer questions like, “Is there a cat or a dog in the image?”
- **Simpler Models and Faster Processing:** Since classification doesn’t need to identify the precise location of objects, it typically requires less processing power and can run faster. This can be ideal for applications where speed is crucial, such as on-device applications like photo sorting or simple image recognition tasks.
- **Applications:** Image classification is often used in medical image analysis (e.g., categorizing disease states from MRI images), product categorization, or content moderation, where understanding the general category is sufficient.

2. Use Object Detection When You Need Precise Locations of Multiple Objects

- **Multiple Objects in Each Image:** Object detection is ideal when you need to locate and identify multiple objects in a single image. This method can answer, “Where are the cat and the dog in the image, and what are their locations?”
- **Bounding Box Localization:** Object detection models generate bounding boxes around each detected object, which is helpful for spatial awareness and applications requiring object positioning, such as detecting vehicles on roads or locating specific products in a warehouse.
- **Complexity and Hardware Requirements:** Object detection generally requires more computational resources than classification. It’s used in applications where accuracy and positioning are crucial, such as surveillance systems, autonomous vehicles, or augmented reality, where knowing both what and where an object is important.

3. Making the Decision Based on Use Case Needs

- If your task only requires knowing *what* is in the image (e.g., a quick classification of a defect in manufacturing), go for image classification.
- If your task involves *knowing both what and where* (such as identifying multiple objects in a crowded scene or tracking items in a retail setting), object detection is the better choice.

Implement custom computer vision models by using Azure AI Vision

Label Images

- **Define Labels:** Create clear, specific labels for each category, like “Apple” or “Banana” in a fruit classification project.
- **Organize by Label:** Group images by their labels for better organization and improved model training accuracy.
- **Use Precise Labels:** Avoid vague terms; use specific names that reflect the image content accurately to enhance model clarity.
- **Ensure Label Consistency:** Apply labels consistently across the dataset to avoid confusion and maintain training accuracy.

Create Custom Vision Resources

- In Azure, create both **Custom Vision Training** and **Prediction** resources.
- Use the Custom Vision web portal to manage these resources.

Start a New Project

- **Project Setup:** Select "New Project" in Custom Vision, provide a project name, and description, and choose the Custom Vision Training resource.
- **Select Classification Type:**
 - **Multilabel** for applying multiple tags per image.
 - **Multiclass** for a single tag per image.
- **Domain Selection:** Choose from options like Generic, Food, Landmarks, Retail, or Compact (for mobile optimization).

Choose Training Images

- **Variety and Format:** Ensure images differ in angle, lighting, etc. Acceptable formats: .jpg, .png, .bmp, or .gif (max size 6 MB).
- **Upload and Tag Images:** Add images by grouping them by tags for easier organization.

Train the Model

- **Initiate Training:** Select the "Train" button. The process identifies each tag’s visual features, which may take several minutes.

Evaluate Model Performance

- **Metrics:** After training, check **Precision** and **Recall** scores under the Performance tab:
 - **Precision:** Fraction of correct classifications.
 - **Recall:** Fraction of actual instances correctly identified.

Publish and Deploy Model

- **Publishing:** After evaluation, publish the model to integrate it with applications.
- **Consumption:** Use REST APIs or SDKs to incorporate the model in image recognition applications.

Analyse images

Select Visual Features for Image Analysis

1. Identify Required Visual Features

- Assess which specific visual elements your application needs, such as detecting objects, assigning tags, analyzing color schemes, or identifying faces.

2. Choose Appropriate Azure AI Vision Capabilities

- Select the most relevant Azure AI Vision options based on these needs. Available features include object detection, tagging, and specialized recognition models for items like faces and logos.

Detect Objects in Images and Generate Descriptive Tags

1. Object Detection

- Detect and locate objects within an image by providing bounding box coordinates and labels for each detected element. Ideal for analyzing scenes with multiple entities.

2. Generate Image Tags

- Use automated tagging to create a descriptive list of content in the image, allowing you to quickly categorize and organize images based on identified items, settings, and activities.

Incorporate Image Analysis Features in Processing Requests

1. Combine Features in API Requests

- Configure API requests to analyze multiple visual features simultaneously, such as object detection, text extraction, and tagging, to gain a thorough analysis from a single request.

2. Specify Analysis Parameters

- Use specific parameters to focus on the desired visual features, enabling efficient data retrieval by requesting only necessary information.

Interpret Responses from Image Processing

1. Review JSON Response Data

- Examine the returned JSON for key details, including recognized tags, detected object locations, confidence levels, and extracted text.

2. Utilize Confidence Scores

- Apply confidence scores to assess the accuracy of detected features, enabling you to filter or prioritize data based on reliability.

Extract Text from Images with Azure AI Vision

1. **Optical Character Recognition (OCR):** Use OCR capabilities to identify and extract text from images, such as printed text on scanned documents, forms, or signs.

2. **Flexible Processing Options:** Choose between synchronous (real-time) or asynchronous (batch) processing modes, allowing you to adapt based on workflow requirements.

Convert Handwritten Text with Azure AI Vision

1. **Handwritten Text OCR:** Enable recognition of handwritten text within images, useful for digitizing notes, forms, or cursive writing.
2. **Embed OCR in Document Workflows:** Incorporate Azure's handwriting OCR within document workflows to easily convert handwritten text into digital formats that can be edited and searched.

Analyze videos

Azure AI Video Indexer

- Uses AI to analyze both audio and video content, generating insights such as tags, entities, and transcripts.
- Enhances media management by automatically creating searchable metadata and closed captions in multiple languages.
- Integrates easily through web portal, widget, or API.
- Supports major compliance standards (e.g., HIPAA, SOC, ISO) for regulated industries.

Azure AI Video Analysis

- Provides tools for object detection, speech recognition, and facial analysis to enhance video content management.
- Generates searchable tags and metadata, improving content discoverability and retrieval.
- Enables enhanced user experience and engagement through features like real-time indexing and transcription.

Azure AI Vision Spatial Analysis

- Detects and tracks objects in live video feeds, useful for real-time monitoring and crowd management.
- Scheduled to retire in March 2025; users are encouraged to transition to Azure AI Video Indexer for expanded features.
- Azure AI Video Indexer offers broader capabilities, including full audio-visual content analysis and cloud integration.

Implement natural language processing solutions

Analyze text by using Azure AI Language

Azure AI-Language is a comprehensive service for building natural language processing (NLP) applications. It enables developers to integrate AI models that can understand, analyze, and generate human language. With Azure AI-Language, you can identify key terms and entities within the text, assess sentiment, summarize content, and create interactive conversational systems. It allows for the training and customization of AI models with minimal machine-learning knowledge, offering a simple yet powerful platform for improving customer interactions and automating text-based workflows.

Key features include:

- **Entity Recognition:** Extract specific entities like people, places, or dates from text.
- **Sentiment Analysis:** Analyze and determine the emotional tone of customer feedback, reviews, or social media posts.
- **Text Summarization:** Automatically summarize lengthy documents or conversations to highlight important points.
- **Conversational Interfaces:** Build and deploy chatbots or virtual assistants to enhance customer engagement.
- **Multilingual Support:** Train models in one language and apply them to multiple languages, making it versatile across different regions.

Extract Key Phrases: Identify and extract important terms and phrases from the text to highlight key topics and concepts.

Extract Entities: Recognize and label specific entities such as names, dates, locations, and organizations from text for better categorization and analysis.

Determine the Sentiment of the Text: Analyze customer feedback, reviews, or conversations to gauge positive, negative, or neutral sentiments, offering insights into customer emotions.

Detect Language Used in Text: Automatically detect the language in which the text is written, enabling support for multiple languages in your applications.

Detect Personally Identifiable Information (PII) in Text: Identify and redact sensitive personal information like names, addresses, and contact details to ensure privacy and comply with data protection regulations.

Process speech by using Azure AI Speech

Text-to-Speech Implementation

- Convert written text into natural-sounding speech.
- Enhances accessibility and user engagement in apps.

Speech-to-Text Implementation

- Transcribe spoken words into text in real time.
- Useful for transcriptions, analysis, and processing of speech data.

Enhance Text-to-Speech with SSML

- Use Speech Synthesis Markup Language (SSML) to control speech features like pitch, speed, and tone.
- Results in more expressive and natural-sounding speech output.

Custom Speech Solutions

- Create specialized speech models to improve recognition accuracy for domain-specific terms.
- Ideal for industries like healthcare, legal, and technical fields.

Intent Recognition

- Analyze user input to determine the underlying intent (e.g., command, query).
- Enables smarter interactions and decision-making in applications.

Keyword Recognition

- Detect specific words or phrases within speech or text.
- Activates commands or identifies key topics in conversations, enhancing voice-driven apps.

Azure AI Translator and Speech Translation Services

Text and Document Translation (Azure AI Translator)

- Translate text and documents across 100+ languages using machine translation.
- Supports instant translation or batch processing for various use cases like multilingual customer support or in-app communication.
- Offers a customizable model for domain-specific terminology, enhancing translation accuracy.

Custom Translation Models

- Train, improve, and publish custom translation models for specific industry needs.
- Leverage AI to adapt translations to the unique terminology and language style required by your application.

Speech-to-Speech Translation (Azure AI Speech)

- Convert spoken language into translated speech using pre-trained voices.
- Enables real-time, multi-lingual conversations without the need to specify the input language.
- Ideal for travel interpreters or business meetings involving multiple languages.

Speech-to-Text Translation (Azure AI Speech)

- Translate speech into text in real time, with language detection and transcription features.
- Use the Speech SDK or CLI for integration, offering accurate transcriptions in target languages.

Multiple Target Language Translation

- Simultaneously translate speech into multiple languages with a single API call.
- Translate input into two or more target languages, helping to reach wider audiences with ease.
- Additional languages can be handled by creating multi-service resources or using separate services for larger-scale translations.

Create a custom question-answering solution by using Azure AI-Language

Azure Conversational Language Understanding (CLU)

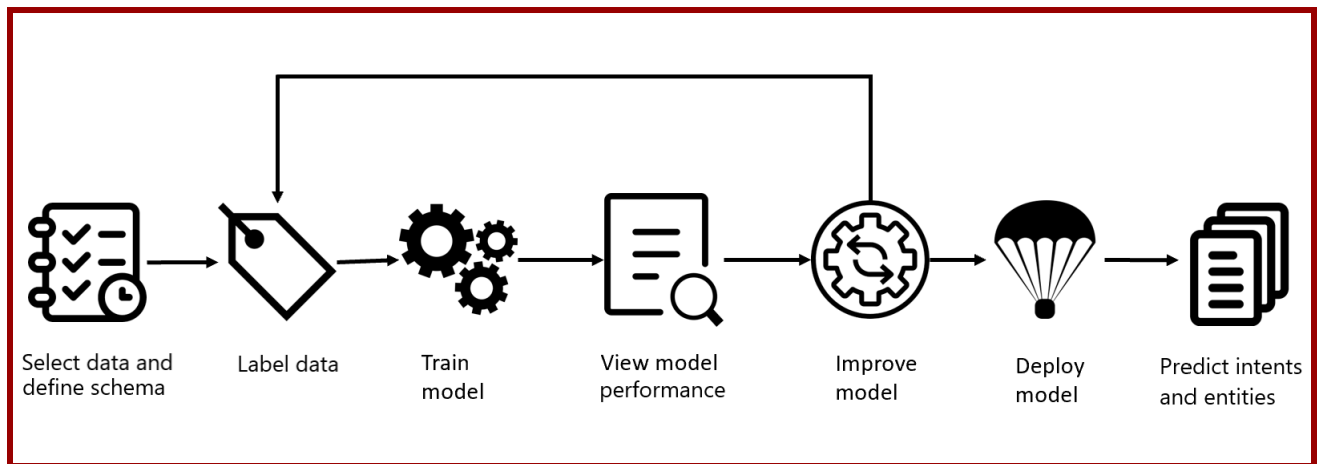
Azure CLU is a cloud-based service that enables developers to create custom natural language understanding (NLU) models for conversational applications. By utilizing machine learning, it allows applications to predict the intent behind user input and extract important data. CLU helps in processing text input, but does not take actions itself. Developers can use a web portal for creating, training, and evaluating models, iterating based on labeled data for improved performance.

Implementations of Azure CLU

1. **Custom Conversational Bots:** CLU powers personalized chatbots for various domains (e.g., retail, food ordering) to interpret user intent and take actions like processing orders.
2. **Human Assistant Bots:** CLU can assist employees in enterprises by routing customer queries to the appropriate support teams or offering HR-related guidance.
3. **Command and Control Applications:** With speech-to-text integration, CLU enables natural language command processing for applications such as media control (e.g., play, pause, or skip songs) or smart devices.
4. **Enterprise Chatbots:** In large organizations, CLU can handle employee queries, manage calendars, and integrate with different services like question answering or feedback collection.

Project Development Lifecycle

1. **Define Schema:** Identify intents (user goals) and entities (key information).
2. **Label Data:** Annotate data to train the model effectively.
3. **Train Model:** Learn from labeled data to improve accuracy.
4. **Evaluate Performance:** Test how the model responds to new inputs.
5. **Improve Model:** Refine the model for better performance.
6. **Deploy Model:** Make the model available via the API.
7. **Predict Intents and Entities:** Use the model to process real-world inputs.



Source: Microsoft Documentation

Knowledge Base - Overview

- **QnA Maker** allows users to create, edit, and manage knowledge bases for automatic question-answering services.
- Knowledge bases are configured through the **QnA Maker portal**, with integration to Azure for resource management.
- Key tasks include adding content from URLs or documents, updating answers, and importing knowledge from existing bases.
- The service supports multi-turn conversations, FAQs, and custom answers for diverse use cases.

Knowledge Base - Characteristics

- **Data Sources:** Knowledge bases are created from various sources like URLs and files, which are organized by data source type.
- **Searchable:** The knowledge base allows users to search questions, answers, or metadata for easy content retrieval.
- **Scalable:** Supports large bases with pagination and search filtering to manage extensive data effectively.
- **Customizable:** Users can add new content, refresh existing URLs, and update settings to fine-tune the knowledge base.

Create a Multi-Language Question Answering Solution

- **Multi-Language Setup:** Set the multi-language option during the creation of the first project to support multiple languages.
- **Create Projects:** For each language, provide the appropriate URL and source language (e.g., German or English) when creating the project.
- **Configure:** Add specific URLs, such as FAQs or support pages, to extract question-answer pairs for each language.
- **Deployment:** After creating the project for each language, deploy it and use the custom question-answering REST API to query the project in the selected language.

Create a Multi-Turn Conversation

- **Multi-Turn Conversations:** Set up conversational AI that can handle multiple interactions with users, maintaining context between queries.
- **Set Up Projects:** Like with the question-answering solution, configure the project with source content, allowing the AI to engage in dialogue across different queries.
- **Manage Dialogue Flow:** Ensure that the system can track context across multiple user inputs, enabling coherent conversations.

Knowledge Mining and Document Intelligence Solutions

Implement an Azure AI Search solution

1. Provision an Azure AI Search Resource

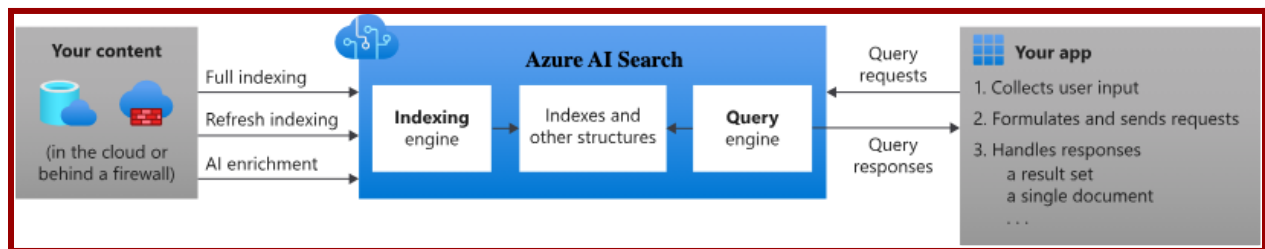
- Create an Azure AI Search service via the Azure portal.
- Choose a tier based on performance and capacity needs.
- Assign a region to optimize search performance and availability.

2. Create Data Sources

- Define the data sources from where content will be ingested.
- Supported data sources include Azure Blob Storage, Cosmos DB, or external databases.
- Use indexers to automate data retrieval and indexing.

3. Create an Index

- Define the structure of searchable content through an index schema.
- Specify fields, data types, and searchable attributes (e.g., text, images).
- Create indexes using the Azure portal or REST APIs.



Source: Microsoft Documentation

4. Define a Skillset

- A skillset enriches indexed data with AI capabilities like text or image analysis.
- Use pre-built or custom AI models for tasks like OCR or sentiment analysis.
- Define a sequence of skills for data enhancement.

5. Implement Custom Skills in a Skillset

- Create custom AI models or processes (e.g., using Azure Machine Learning) to process data.
- Integrate these custom skills into a skillset for indexing.

6. Create and Run an Indexer

- Indexers automate the process of transforming and ingesting raw data into a searchable index.
- Define the data source and schema for the indexer.
- Schedule indexing to keep data up to date.

7. Query an Index

- Use queries to retrieve data from the index.
- Syntax: Support for full-text search, vector search, fuzzy search, wildcard, and more.
- Include filters and sorting to refine search results based on relevance.

8. Manage Knowledge Store Projections

- Projections are configurations for how data is retrieved and displayed in a Knowledge Store.
- Types of projections include file, object, and table projections, based on content types.
- Manage these projections for specific search or analytic needs.

Implement an Azure AI Document Intelligence solution

1. Setting Up a Document Intelligence Resource

- Navigate to the Azure portal and create a new resource.
- Search for "Document Intelligence" and click "Create".
- Provide necessary details, such as subscription, resource group, region, and pricing tier (the free tier is available for testing).
- After validation, finalize the setup and complete the deployment.

2. Leveraging Prebuilt Models for Document Data Extraction

- Use Azure's prebuilt models to automatically extract essential data such as key-value pairs, tables, and text from documents.

3. Developing a Custom Document Intelligence Model

- Build a model suited to your specific needs by using Azure's training capabilities.
- Train the model using your sample documents and apply custom logic for precise data extraction.

4. Training, Testing, and Deploying a Custom Document Model

- Train the custom model with your document samples, ensuring the model adapts to your data.
- Test the model for accuracy, and then publish it for use in applications.

5. Creating a Combined Document Intelligence Model

- Merge multiple models or techniques to create a unified model that handles more complex document processing tasks.

6. Integrating Document Intelligence into Azure AI Search as a Custom Skill

- Add your custom model to Azure AI Search as a skill, which enhances search functionalities and enables more effective document retrieval and analysis.

Implement generative AI solutions

Use Azure OpenAI Service to generate content

- **Azure OpenAI Service** provides access to advanced models like GPT-4 and GPT-3.5-Turbo.
- Models can be accessed through **REST APIs**, **Python SDK**, or **Azure AI Studio**.
- Use cases include **content generation**, **text summarization**, **semantic search**, and **natural language-to-code translation**.
- Integrates with **Azure**, offering robust **security** and **management tools**.
- Enables **efficient deployment** and **customization** of AI models for various applications.

Provision an Azure OpenAI Resource:

Create a new Azure OpenAI resource in the Azure portal or via CLI/PowerShell by following the resource setup process.

Select and Deploy an Azure OpenAI Model:

Choose a model (like GPT-4, GPT-3.5, or DALL-E) and deploy it through the Azure portal or APIs for generating text or images.

Submit Prompts to Generate Natural Language:

Input text-based prompts and receive natural language responses, such as content generation or summarization, using the completions API.

Submit Prompts to Generate Code:

Use prompt-based requests to generate code (e.g., Python, JavaScript) via models like Codex and GPT series.

Use the DALL-E Model to Generate Images:

Provide text prompts to DALL-E for image creation from descriptions, including variations and adjustments to the generated content.

Use Azure OpenAI APIs to Submit Prompts and Receive Responses:

Interact with models using REST APIs or SDKs to submit requests and process generated content.

Use Large Multimodal Models in Azure OpenAI:

Leverage powerful models like GPT-4 with vision capabilities to process and analyze images and text in a unified API workflow.

Optimize Generative AI

To optimize Generative AI for Independent Software Vendors (ISVs) and Digital Natives, understanding and managing the inherent limits and quotas of OpenAI models is crucial. These limits ensure consistent and reliable performance across all users. To maximize the effectiveness of these models, it is important to:

- **Configure parameters for generative behavior:**
 - Adjust parameters like *temperature* to control response creativity (higher values = more diverse, lower values = more deterministic).
 - Set *max tokens* to define the length of the generated output.
 - Use *stop sequences* to specify where the generation should end, helping manage content length.
- **Apply prompt engineering techniques:**
 - Refine input prompts to guide model outputs (e.g., using specific contexts like "Summarize AI for beginners").
 - Experiment with phrasing to improve the accuracy and relevance of responses, ensuring alignment with your specific needs.
- **Use your own data with an Azure OpenAI model:**
 - Integrate your proprietary data to customize the model's responses for business-specific tasks.
 - Enhance the model's ability to understand and generate contextually relevant content using domain-specific information.
- **Fine-tune an Azure OpenAI model:**
 - Retrain a pre-existing model on your specific dataset to improve task-specific performance.
 - Optimize models like GPT-4 for specialized applications, such as customer support or domain-specific question answering.